

Supplementary Material 5: Generating the 2NC7 data

This data is constructed by simulating two clusters, each consisting of 1000 observations. The C-shaped cluster is generated from $\theta \sim U(-3\pi/2, 0)$, $X_1 = \sin(\theta)$, $X_2 \sim U(0, 2)$ (adding thickness to the C), $X_3 = \text{sign}(\theta) \times (\cos(\theta) - 1)$, $X_4 = \cos(\theta)$. Observations lie on a 2- D manifold in 7- D . The other cluster is from $X_1 \sim U(0, 2)$, $X_2 \sim U(0, 3)$, $X_3 = -(X_1^3 + X_2)$, and $X_4 \sim U(0, 2)$. It is also curved, but observations lie on a 3- D manifold in 7- D . Three more variables, X_5, X_6, X_7 , that are small amounts of pure noise are added. We would consider $T = (X_1, X_2, X_3, X_4)$ to be the geometric structure (true model) that we hope to capture.

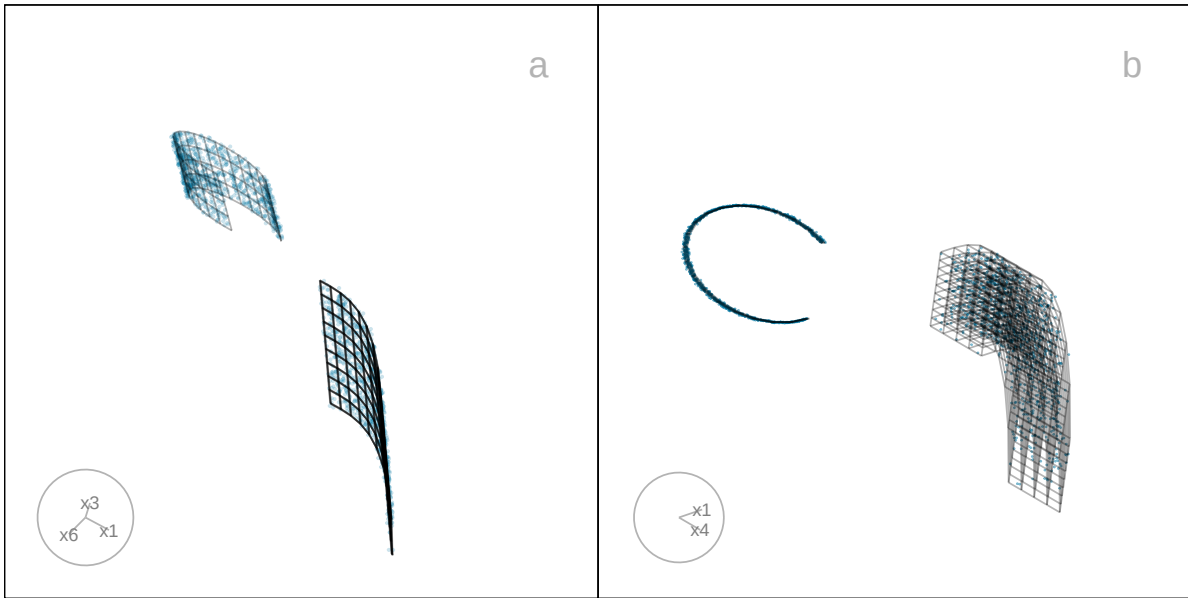


Figure 1: Two projections of the p - D true model overlaying the data are shown in a, b. Video of the langevitour animations is available at <https://youtu.be/35TrnYJsUUI>.